Hematology in Critical Care – Chapter Summary

Transfusion • Coagulopathy • Thrombosis • TMA

ICU / EDIC Board Review • MCQ-Oriented

1. Transfusion Thresholds

- **Restrictive Hb < 7 g/dL** in most ICU patients (TRICC, TRISS)
- **Acute MI:** Hb < 8 (liberal < 10 possibly better – MINT)
- **Cardiac surgery:** < 7.5 (TRICS III)
- **Brain injury:** liberal (< 9–10) may help (TRAIN)
- **Platelets:** < 10k prophylactic; < 50k bleeding / surgery; < 100k neurosurgery / intracranial bleed; CVC ok > 10k (PACER)
- **FFP:** bleeding with INR > 1.5–2; not for prophylactic INR correction
- **Cryoprecipitate:** fibrinogen < 1.5 g/L (< 2 in obstetrics)

2. Transfusion Reactions

Reaction	Features
Acute hemolytic (ABO)	Fever, hypotension, hemoglobinuria, DIC – minutes
TRALI	≤ 6 h, hypoxemia, bilateral infiltrates, hypotension, fever; donor anti-HLA/HNA
TACO	Hypertension, ↑ BNP, fluid overload → diuretics
Febrile non-hemolytic	Fever, chills – cytokines
Anaphylaxis	IgA deficiency (anti-IgA)
Delayed hemolytic	3–14 days, ↓ Hb, ↑ bilirubin
TA-GVHD	Irradiated products prevent

- **Bacterial contamination:** platelets (room temperature storage)
- **Massive transfusion:** hypocalcemia, hyperkalemia, hypothermia, acidosis, dilutional coagulopathy

3. Massive Hemorrhage

- **Ratio 1:1:1** (PROPPR – ↓ deaths from exsanguination)
- **TXA 1 g + 1 g** within 3 h (CRASH-2)
- **Viscoelastic (ROTEM / TEG)**-guided therapy
- Targets: fibrinogen > 1.5–2, plt > 50k, iCa > 1.1, temp > 35 °C, pH > 7.2
- **Lethal triad:** hypothermia, acidosis, coagulopathy (+ hypocalcemia = diamond)
- **rFVIIa:** off-label, thrombosis risk

4. ROTEM / TEG Interpretation

Parameter	Abnormality → Treatment
↑ CT / R time	Factor deficiency → FFP / PCC
↓ A10 FIBTEM / ↓ angle	Low fibrinogen → cryo / fibrinogen
↓ A10 EXTEM, normal FIBTEM / ↓ MA	Low platelets → platelets
↑ LI30 / ↑ LY30	Hyperfibrinolysis → TXA
CT INTEM ↑, HEPTEM normal	Heparin effect → protamine

5. DIC

- **ISTH score ≥ 5:** platelets, D-dimer / FDP, PT, fibrinogen
- **Sepsis-induced coagulopathy (SIC)** score: platelets, INR, SOFA
- Rx: **treat cause**; products only if bleeding / procedure
- Thrombotic-dominant (sepsis) vs fibrinolytic (APL, trauma, obstetric)
- Heparin for thromboembolic predominant DIC

6. Thrombocytopenia in ICU

- Common causes: sepsis, DIC, drugs, dilution, HIT, TMA, liver disease / hypersplenism, ITP
- **Pseudothrombocytopenia:** EDTA clumping → citrate tube
- **Drug-induced:** heparin, linezolid, vancomycin, β-lactams, GPIIb/IIIa inhibitors
- **ITP:** IVIG + steroids; TPO agonists
- **Post-transfusion purpura:** 7–10 days after, anti-HPA-1a → IVIG

7. Heparin-Induced Thrombocytopenia

- **4T score:** Thrombocytopenia (> 50% fall), Timing (5–10 days), Thrombosis, oTher cause
- Low 4T (≤ 3) → excludes; intermediate / high → anti-PF4 ELISA ± functional (SRA / HIPA)
- **Stop all heparin** (incl. flushes) → non-heparin anticoagulant: **argatroban** (hepatic), bivalirudin, fondaparinux, DOAC
- **No platelet transfusion**; **no warfarin** until plt > 150k (venous limb gangrene)
- **VITT** (vaccine-induced): similar, anti-PF4, CVST → IVIG + non-heparin anticoagulant

8. Thrombotic Microangiopathies

- **MAHA** (schistocytes, ↑ LDH, ↓ haptoglobin) + thrombocytopenia + organ damage

TMA	Key feature	Rx
TTP	ADAMTS13 < 10%, neuro, PLASMIC score	**Plasma exchange** + steroids + caplacizumab ± rituximab
STEC-HUS	Bloody diarrhea, E. coli O157	Supportive, no antibiotics
Atypical HUS	Complement dysregulation, AKI	**Eculizumab / ravulizumab**
DIC	↑ PT, ↓ fibrinogen	Treat cause
HELLP / malignant HTN / scleroderma / drugs	Context	Treat cause

- **Platelet transfusion** in TTP: avoid unless life-threatening bleeding

9. Anticoagulant Reversal

Agent	Reversal
Warfarin	4F-PCC 25–50 U/kg + vitamin K 10 mg IV
UFH	Protamine 1 mg / 100 U (last 2–3 h)
LMWH	Protamine partial (~60%)
Dabigatran	Idarucizumab 5 g; dialysis
Rivaroxaban / apixaban	Andexanet alfa or 4F-PCC 50 U/kg
Fondaparinux	No specific (rFVIIa)
Thrombolytics	Cryoprecipitate, TXA

- Antiplatelets: platelets not routinely (PATCH harm in ICH); desmopressin for uremic platelets

10. VTE Prophylaxis & Treatment

- **LMWH** preferred prophylaxis (PROTECT: dalteparin = UFH, ↓ PE)
- **Mechanical:** IPC add-on no benefit (PREVENT) – use if pharmacologic contraindicated
- **IVC filter:** only if anticoagulation contraindicated with acute VTE
- **Catheter-related thrombosis:** anticoagulate ≥ 3 months
- **Cancer VTE:** LMWH or DOAC (apixaban)

11. Hemoglobinopathies & Anemia

- **Sickle cell acute chest syndrome:** new infiltrate + fever / chest pain / hypoxia → O_2, antibiotics, incentive spirometry, **exchange transfusion** (target HbS < 30%)
- **Hyperhemolysis** after transfusion
- **Avoid:** hypoxia, dehydration, hypothermia, acidosis
- **G6PD deficiency:** oxidant drugs (primaquine, dapsone, rasburicase, methylene blue) → hemolysis
- **Autoimmune hemolysis:** warm (IgG, steroids) / cold (IgM, avoid cold)
- **Methemoglobinemia:** SpO_2 ~85%, chocolate blood → methylene blue 1–2 mg/kg (not in G6PD)

12. Anemia of Critical Illness

- Inflammation (hepcidin), blood sampling, bleeding, ↓ EPO
- Small-volume tubes ↓ phlebotomy loss
- **EPO:** no benefit routinely; ↑ thrombosis
- **IV iron:** no clear outcome benefit in ICU
- Jehovah's Witness: minimize sampling, cell salvage, EPO, iron, TXA

13. Hematologic Malignancy Emergencies

- **APL:** DIC with hyperfibrinolysis → ATRA immediately, cryo / platelets
- **Hyperviscosity** (Waldenström IgM, myeloma): neuro, visual, bleeding → plasmapheresis
- **Leukostasis:** WBC > 100k → leukapheresis, hydroxyurea, avoid RBC transfusion
- **HLH:** see rheumatology

14. High-Yield MCQ Pearls

- ☑ Restrictive Hb < 7 in most ICU patients
- ☑ Massive transfusion 1:1:1 + TXA ≤ 3 h
- ☑ TRALI: hypotension + normal BNP; TACO: HTN + ↑ BNP
- ☑ IgA deficiency → anaphylaxis
- ☑ HIT: stop heparin → argatroban; no warfarin, no platelets
- ☑ TTP: ADAMTS13 < 10% → plasma exchange
- ☑ aHUS → eculizumab
- ☑ Warfarin reversal: 4F-PCC + vitamin K
- ☑ Dabigatran → idarucizumab
- ☑ Sickle chest syndrome → exchange transfusion
- ☑ Methylene blue contraindicated in G6PD
- ☑ APL → give ATRA immediately

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